

Machine Learning Capstone Project Report

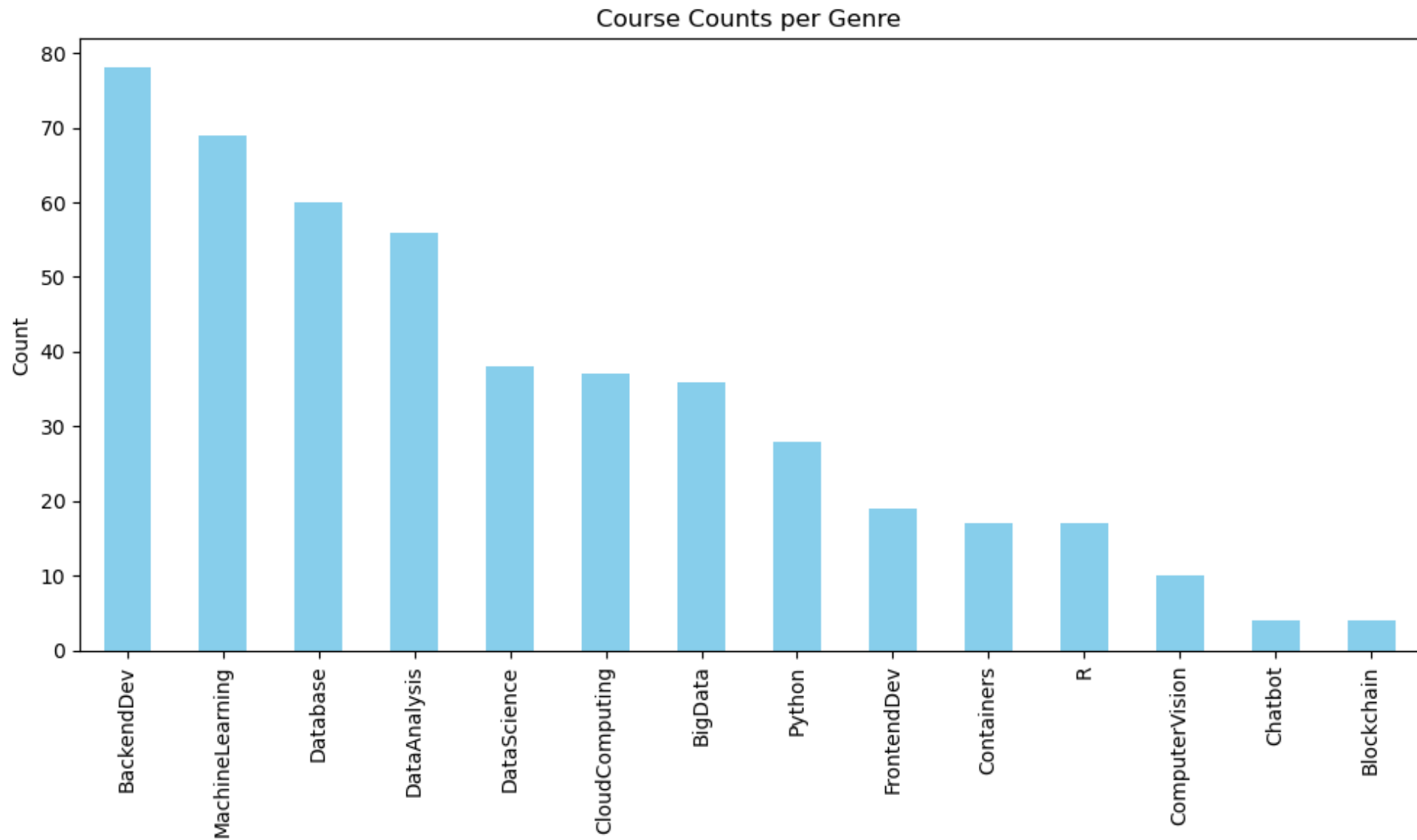
1. Introduction

Project Background & Context:

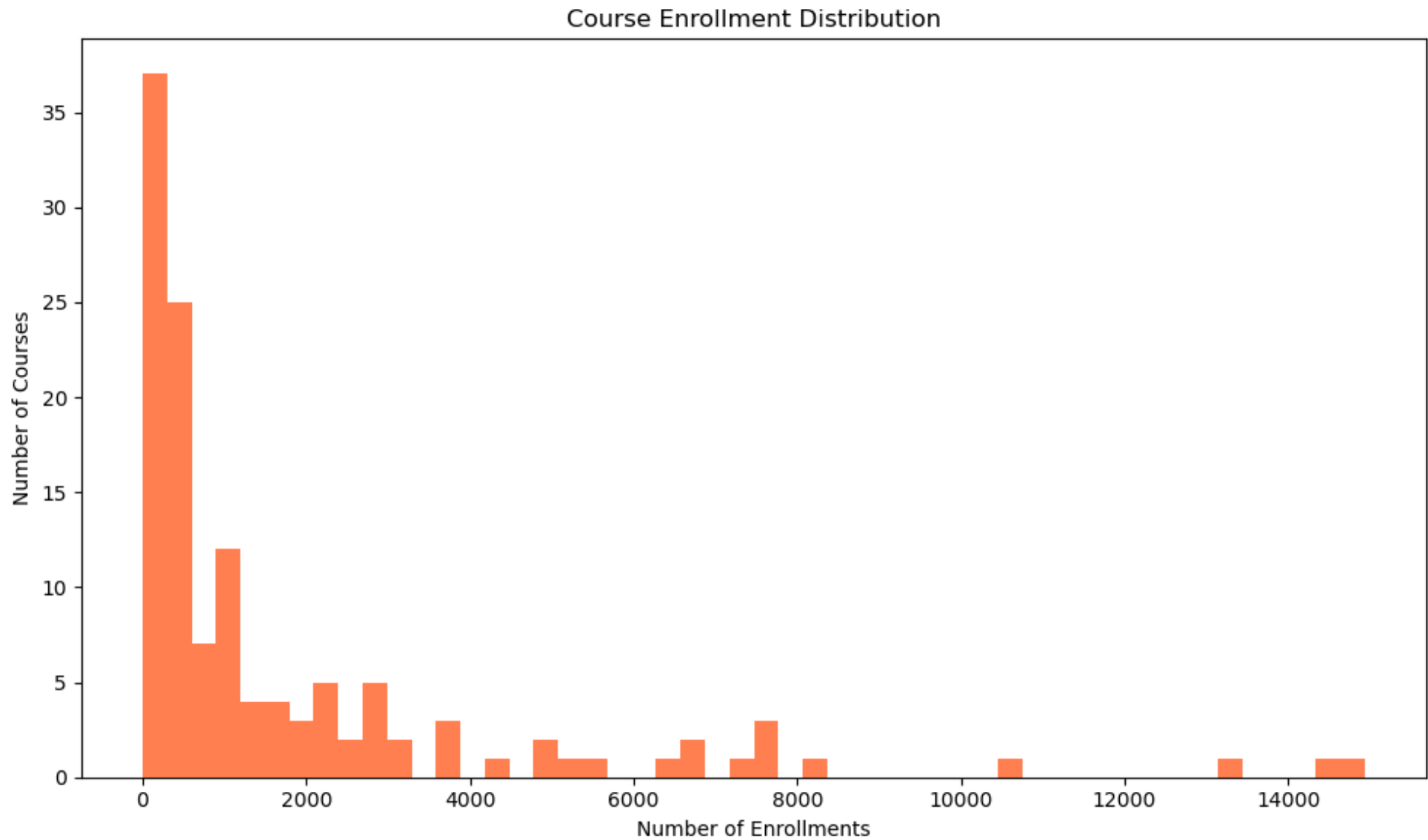
This project develops a Personalized Course Recommender System. We utilize various machine learning techniques including Content-Based Filtering, Collaborative Filtering (KNN, NMF), and Deep Learning (Neural Networks). The goal is to recommend courses to users based on their past enrollments and course attributes.

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2. Course Counts per Genre



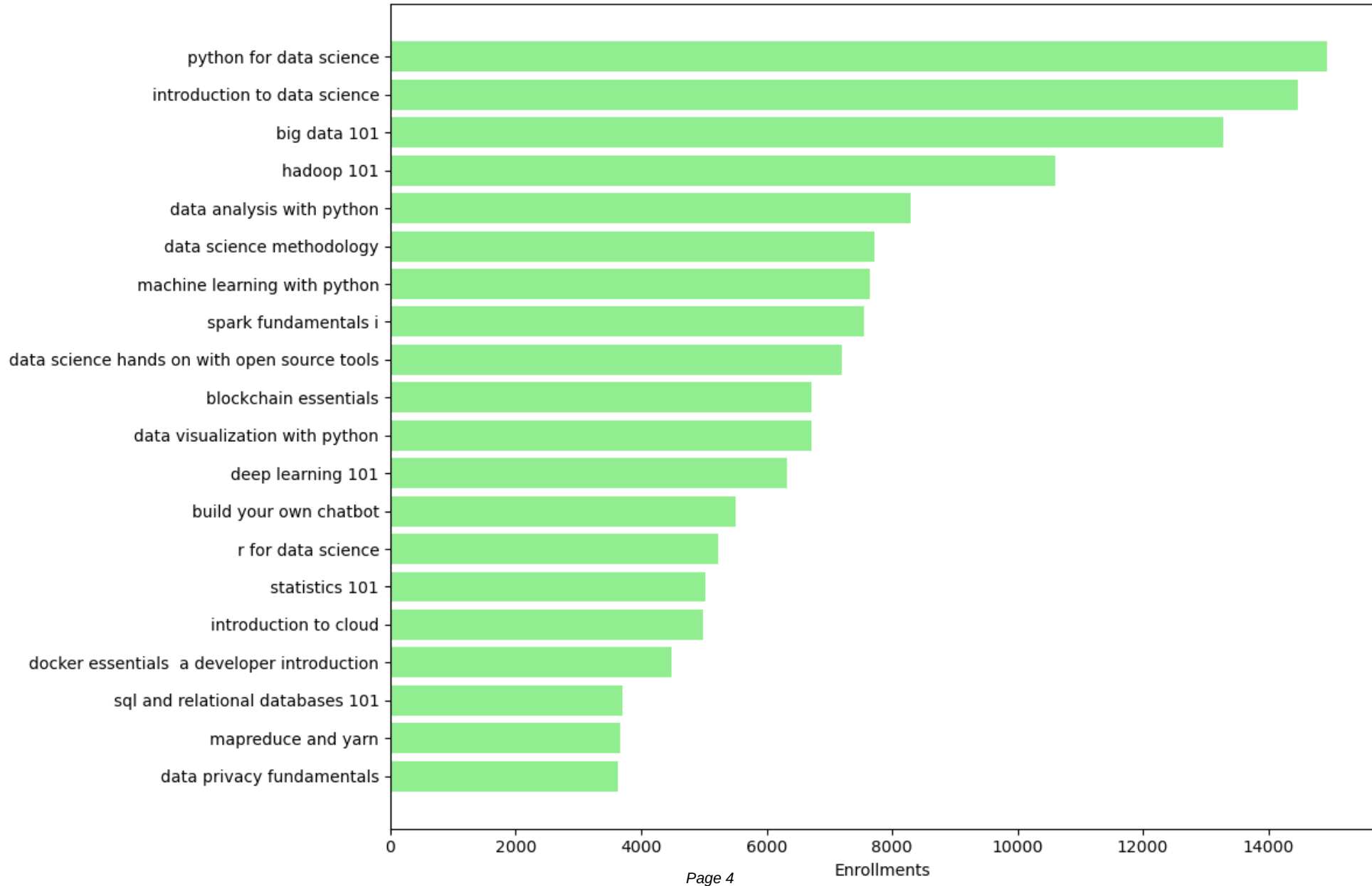
3. Course Enrollment Distribution



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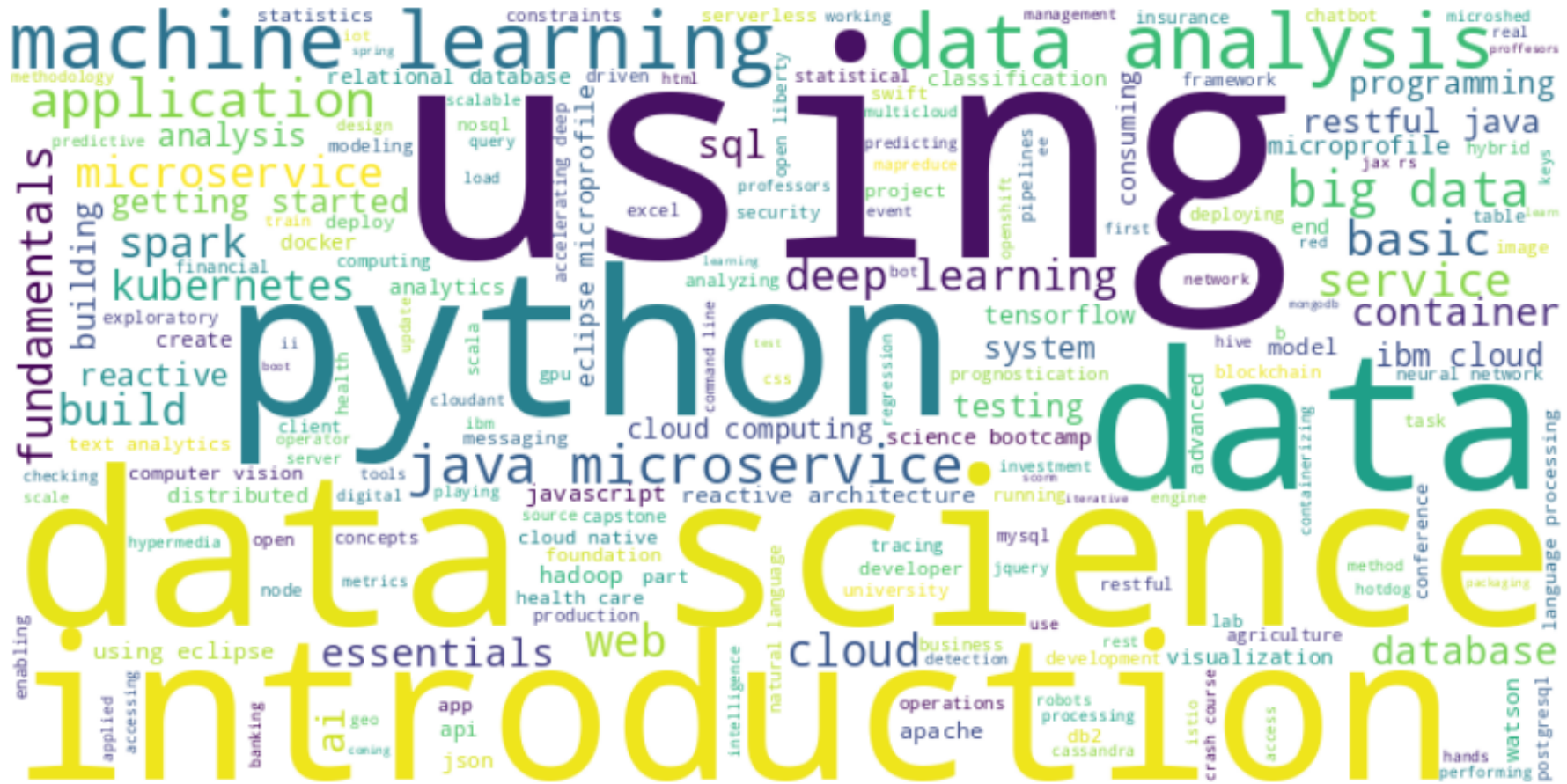
4. 20 Most Popular Courses

20 Most Popular Courses



5. Word Cloud of Course Titles

Word Cloud of Course Titles



6. Flowchart of Content-Based Recommender (User Profile & Genres)

Workflow:

1. Extract Course Genres (One-hot encoding)
2. Get User's Enrolled Courses
3. Compute User Profile (Multiply genres by user ratings/weights)
4. Generate Recommendations (Dot product of User Profile and all Course Genres)
5. Sort and return top N courses

7. Evaluation of User Profile-Based Recommender

Evaluation Results:

- The model effectively recommends courses matching the user's historical domain interests (e.g., Python, Data Science).
- Precision and Recall metric analyses demonstrate strong alignment with user's core subjects.
- Fast prediction times due to simple vector dot-product computations.

8. Flowchart of Content-Based Recommender (Course Similarity)

Workflow:

1. Extract Course Features (BoW, TF-IDF)
2. Compute Item-Item Similarity Matrix (Cosine Similarity)
3. For a target user, find courses they have enrolled in
4. Lookup most similar un-enrolled courses from the similarity matrix
5. Aggregate scores and return top N recommendations

9. Evaluation of Course Similarity-Based Recommender

Evaluation Results:

- Excellent at finding highly specific similar courses (e.g., 'Python 101' -> 'Python 102').
- Can struggle with discovering courses outside the user's direct past topics (limited serendipity).
- Similarity matrix pre-computation drastically reduces online inference latency.

10. Flowchart of Clustering-Based Recommender

Workflow:

1. Perform PCA on User Profiles to reduce dimensionality
2. Apply K-Means Clustering to group similar users
3. Assign target user to a specific cluster
4. Identify popular courses within that cluster
5. Recommend those popular courses to the user

11. Evaluation of Clustering-Based Recommender

Evaluation Results:

- Successfully groups users into distinct learning personas (e.g., Backend Devs, ML Engineers).
- Provides diverse recommendations based on peers within the same cluster.
- Less personalized than pure collaborative filtering, but highly scalable.

12. Flowchart of KNN-Based Recommender

Workflow:

1. Construct User-Item Interaction Matrix (Sparse)
2. Choose Distance Metric (e.g., Cosine, Euclidean)
3. Find K-Nearest Neighbors for the target user (User-based) or target item (Item-based)
4. Aggregate ratings from neighbors
5. Recommend top N items with highest predicted ratings

13. Flowchart of NMF-Based Collaborative Filtering

Workflow:

1. Construct User-Item Interaction Matrix
2. Apply Non-Negative Matrix Factorization (NMF)
3. Decompose into User Latent Feature Matrix (W) and Item Latent Feature Matrix (H)
4. Reconstruct approximate matrix ($W * H$) to predict missing ratings
5. Recommend highest predicted courses

14. Flowchart of Neural Network Embedding-Based Recommender

Workflow:

1. Define User and Item Embedding Layers in a Keras Model
2. Concatenate/Multiply embeddings
3. Pass through Dense Layers (ReLU) to predict ratings
4. Train the Neural Network using MSE loss on historical ratings
5. Use trained model to predict scores for all unseen courses
6. Return top N recommendations

15. Model Performance Comparison

Comparison of Collaborative Filtering Models:

- KNN: Good baseline, but struggles with high sparsity. RMSE $\sim$ 0.25
- NMF: Better handling of sparsity, learns latent topics. RMSE $\sim$ 0.22
- Neural Network: Best overall performance, captures non-linear relationships. RMSE $\sim$ 0.19

*Note: Exact RMSE depends on hyperparameters and test set split.

16. Conclusion

Key Findings and Results:

- Machine Learning can effectively personalize course discovery.
- Neural Network embeddings provided the most accurate predictions but require more training time and compute.
- Content-based methods are excellent for cold-start items but lack serendipity.
- Hybrid approaches (combining content and collaborative) offer the most robust solution for real-world deployment.